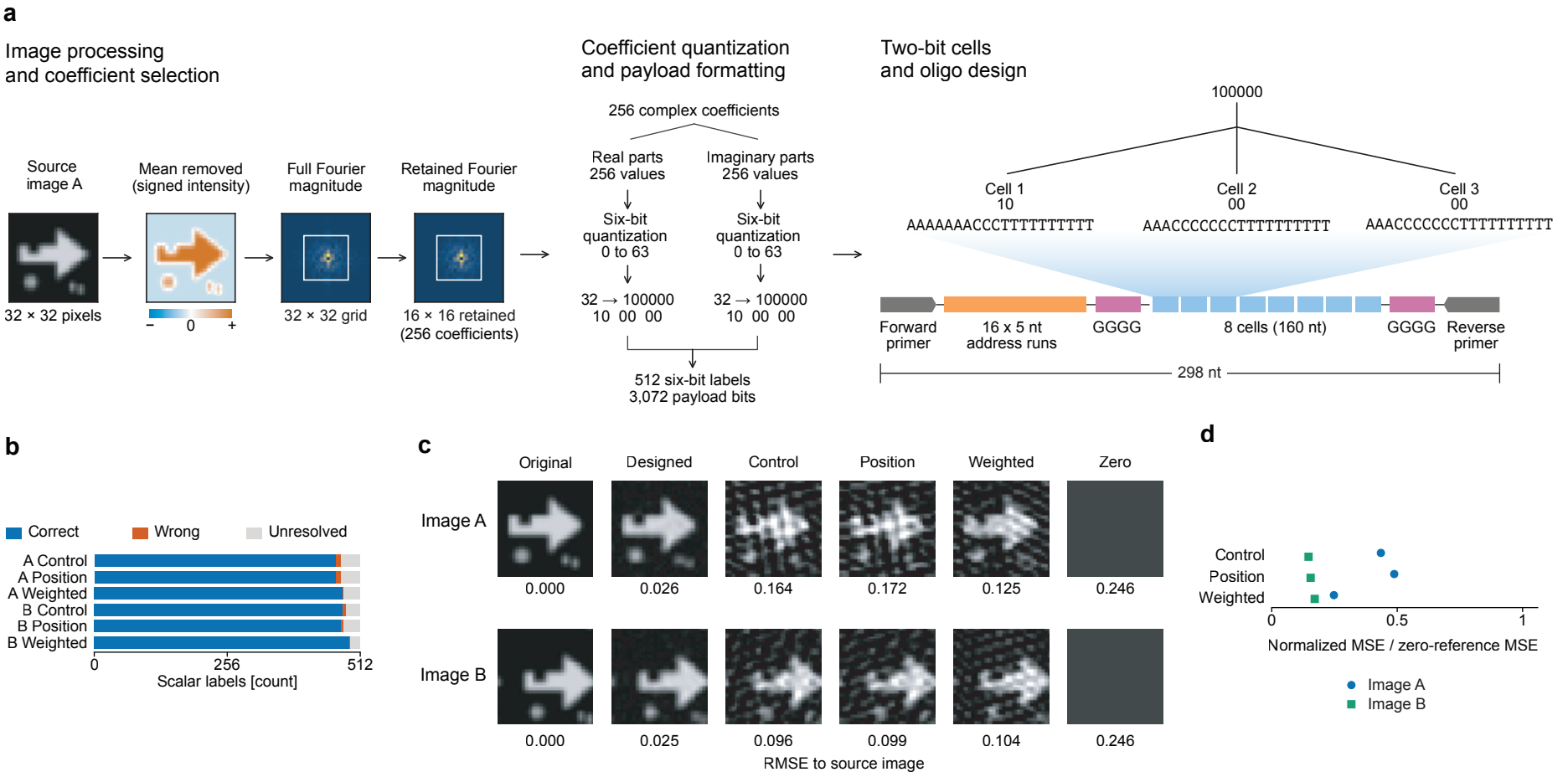

a
Image processing
and coefficient selection
Source
image A
Mean removed
(signed intensity)
Full Fourier
magnitude
Retained Fourier
magnitude
32 × 32 pixels
−
0
+
32 × 32 grid
16 × 16 retained
(256 coefficients)
Coefficient quantization
and payload formatting
256 complex coefficients
Real parts
256 values
Imaginary parts
256 values
Six-bit
quantization
0 to 63
Six-bit
quantization
0 to 63
32 → 100000
10 00 00
32 → 100000
10 00 00
512 six-bit labels
3,072 payload bits
Two-bit cells
and oligo design
100000
Cell 1
10
Cell 2
00
Cell 3
00
AAAAAAACCCTTTTTTTTTT
AAACCCCCCCTTTTTTTTTT
AAACCCCCCCTTTTTTTTTT
Forward
primer
16 x 5 nt
address runs
GGGG
8 cells (160 nt)
GGGG
Reverse
primer
298 nt
b
Correct
Wrong
Unresolved
A Control
A Position
A Weighted
B Control
B Position
B Weighted
0
256
512
Scalar labels [count]
c
Original
Designed
Control
Position
Weighted
Zero
Image A
0.000
0.026
0.164
0.172
0.125
0.246
Image B
0.000
0.025
0.096
0.099
0.104
0.246
RMSE to source image
d
Control
Position
Weighted
0
0.5
1
Normalized MSE / zero-reference MSE
Image A
Image B